5 **(a)** State the word equation representing the first law of thermodynamics.

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..... [1]

(b) Fig. 5.1 shows the *P*-*V* graph of a system of ideal gas undergoing three different processes.

Along the path **ABC** the work done by the gas is 40 kJ and the heat energy supplied to the gas is 90 kJ.

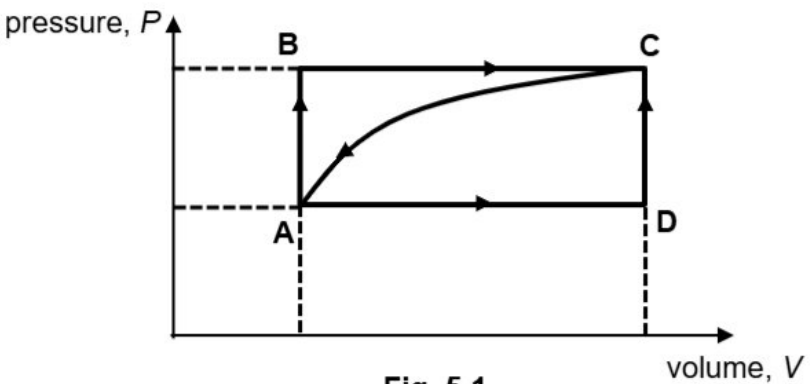


Fig. 5.1

(i) Calculate the internal energy at **C** if the internal energy at **A** is 45 kJ.

internal energy = ... kJ [2]

(ii) Along the path **ADC** the work done by the gas is 20 kJ.

Calculate the heat energy supplied to the gas along the path **ADC**.

energy supplied = ... kJ [2]

(iii) The work done on the gas when it returns along the curved path **CA** is 31 kJ.

1. Calculate the net work done on the gas in the cycle **ABCA**.

net work done on gas = ... kJ [2]

2. Calculate the net heat supplied to the gas in the cycle **ABCA**.

Explain your answer clearly.

.....

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net heat supplied = ... kJ [2]